Zulfiya Usmonova

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Research Interests: AI Evaluation & Benchmarking — LLM Robustness — AI Safety — Human-AI Interaction

Education

Technical University of Munich (TUM) Munich, Germany
MSc in AI in Society, GPA: 1.3 *Oct. 2025 – March. 2027*

- Coursework: Human-AI Interaction, Foundations of AI and Data Science, AI in Metaverse, Introduction to Deep Learning, Path to AGI, Augmented Human: Exploring the Future of Human-Technology Symbiosis.

New Uzbekistan University Tashkent, Uzbekistan
BSc in Software Engineering, GPA: 4.1/4.5 (Rank: 6/115) *Oct. 2021 – Jun. 2025*

- Honor of University Award (2023) and Top Student Award (2022).
- Engaged in Topological Data Analysis seminars and guest lectures from TUM, Cornell, and MIT faculty.

Comenius University in Bratislava Bratislava, Slovakia
Exchange Semester (NSP Scholarship Recipient) *Feb. 2025 – Jun. 2025*

- Actively volunteered in conferences and seminars hosted by AI industry professionals (e.g., Google DeepMind founders, Student Conferences).
- Coursework: Neural Networks, Statistical Methods in AI, Combinatorics & Graph Theory, Discrete Math 2, Biostatistics

Publications & Academic Writing

NAACL 2027 submission (first author) March. 2026 – Oct. 2026
Supervised by Prof. Enkelejda Kasneci *TUM, Germany*

- Designed and implemented reproducible experiments to evaluate information preservation and robustness across open-weight LLMs.
- Built evaluation pipelines combining embedding-based semantic similarity, statistical analysis, predictive modeling
- Studied semantic drift phenomena in LLM transmission chains.

Book Chapter (first author) Dec. 2023 – Apr. 2024
Supervised by Prof. M. U. Ayoobkhan *Tashkent, Uzbekistan*

- Cloud Computing for Smart Education, Transformative Pedagogy: Collaborative Learning with Cloud Architecture and Virtual Programming Labs; CRC Press, Taylor&Francis Group.
- Abstract available: https://bit.ly/book-chapter-link

Research & Work Experience

Student Research Assistant Nov. 2025 – Present
Chair for Human Centered Technologies for Learning *TUM, Germany*

- Designing and implementing a prompted weak supervision pipeline using local large language models for automated teaching quality assessment.
- Prototyping and benchmarking local open-weight LLMs and setting up server-based experimentation workflows and lightweight model-testing pipelines for scalable and reproducible evaluation.
- Reviewed and synthesized 150 papers on personalized AI coaching agents in immersive learning environments, contributing to manuscript writing and figure preparation.
- Second author of a systematic review submitted to *Computers & Education: Artificial Intelligence*.

TUM Social AI Club - AI Engineer Oct. 2025 – Present
Collaboration with InitiativGruppe and UN Women Germany *Munich, Germany*

- Built and deployed an LLM-based chatbot for a public-facing website, implementing safety-oriented prompting strategies and response validation for reliable user interaction.
- Leading the WRPBench team in collaboration with UN Women Germany, coordinating benchmark design, prompt development, and expert review for women's rights and gender-equality scenarios.
- Developing source-grounded prompt generation and evaluation workflows to assess LLM factual accuracy, safety, stereotype resistance, framing fairness, and consistency.

AI Research / Engineering Project Nov. 2025 – Jan. 2026

Multi-Agent LLM Backend for Metaverse Simulation *TUM, Germany*

- Designed and implemented a Python Flask backend linking a Godot-based simulation with locally hosted LLMs, enabling heterogeneous multi-agent architectures with memory for autonomous decision-making in a shared virtual environment.
- Debugged and stabilized the system under concurrent requests, addressing timeouts, model latency, and API incompatibilities to ensure reliable agent execution.
- Authored comprehensive technical documentation covering system architecture, API integration, setup, deployment, and troubleshooting to support reproducibility and future development.

Bachelor's Thesis Sep. 2024 – Apr. 2025

When AIs Change Their Minds: Testing LLM Robustness to Altered Voting Protocols *Tashkent, Uzbekistan*

- Built a 100-agent simulation across 10 LLM families to test robustness under altered voting protocols.
- Designed a two-stage study: (i) individual baseline establishes per-model agree/disagree rates; (ii) network stage embeds agents in random d-regular graphs ($d = 1 - 9$) and measures vote-flip rates as peer exposure grows.
- Discovered that modest network degree ($d > 1$) can invert the majority for smaller instruction-tuned models (phi, Gemma) while larger, alignment-optimized models (GPT-4, Gemini) remain stable.
- Code and report: https://github.com/UsmonovaZulfiya/bachelor-s-thesis.git

Skills

Programming: Python, SQL, R
Research & Evaluation: experimental design, benchmark development, robustness testing, statistical analysis
ML & Data: PyTorch, Hugging Face Transformers, scikit-learn, Pandas, Matplotlib, NetworkX
LLMs & Agents: local model serving, embeddings, tool calling, agent memory, multi-agent orchestration (LangGraph, AutoGen)
Engineering & Compute: Flask APIs, concurrent request handling, structured JSON interfaces, Git, Slurm
Simulation & Writing: Unity, Godot, LaTeX

Selected Competitions and Volunteering

Competitions & Hackathons: Huawei Seeds for the Future Global Competition (2nd place), NASA Space Apps Challenge Uzbekistan (Winner)

Volunteering: Women In Tech Uzbekistan Chapter Marketing Team Lead (2025–Present); Huawei ICT Ambassador (2024–25); TechnovationGirls Mentor and Ambassador (2022–24); Academic Mentor at Summer School (2025); Education Minister & General Secretary, NewUU Student Council (2022–24).